| | |
|---|---|
| United States Patent | 12404802 |
| Kind Code | B2 |
| Date of Patent | September 02, 2025 |
| Inventor(s) | Gastaldi; Patrick |


## Combustion chamber for high performance H2 direct injection engine


### Abstract

A hydrogen direct injection engine includes a cylinder and a combustion chamber positioned at a center of the cylinder, where an axis of the combustion chamber is approximately identical to an axis of the cylinder. The hydrogen direct injection engine also includes one or more intake valves positioned around the cylinder, one or more exhaust valves positioned around the cylinder, a cylinder head mounted across a top surface of the cylinder, and a fuel injector positioned vertically at the center of the cylinder. The hydrogen direct injection engine further includes a piston disposed within the cylinder, and a spark plug positioned between the one or more intake valves.



**Inventors:** **Gastaldi; Patrick (Rueil-Malmaison, FR)**

**Applicant:** **SAUDI ARABIAN OIL COMPANY** (Dhahran, SA)



**Family ID:** **91302292**

**Assignee:** **SAUDI ARABIAN OIL COMPANY (Dhahran, SA)**

**Appl. No.:** **18/299802**

**Filed:** **April 13, 2023**

### Prior Publication Data

| Document Identifier | Publication Date |
|---|---|
| US 20240344479 A1 | Oct. 17, 2024 |

### Publication Classification

**Int. Cl.:** **F02B43/12** (20060101); **F02B23/10** (20060101); F02B75/12 (20060101)

**U.S. Cl.:**

CPC **F02B43/12** (20130101); **F02B23/101** (20130101); F02B2023/102 (20130101);

F02B2075/125 (20130101)

## Field of Classification Search

**CPC:** F02B (23/101); F02B (43/12)

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC |
|---|---|---|---|---|
| 4329949 | 12/1981 | Abts | 123/90.34 | F02F 1/38 |
| 5645023 | 12/1996 | Regueiro | 123/90.22 | F01L 1/14 |
| 6260518 | 12/2000 | Jingu | N/A | N/A |
| 6947830 | 12/2004 | Froloff | 701/111 | F02D 41/3076 |
| 6951193 | 12/2004 | Draper | 123/41.17 | F01P 1/02 |
| 7526944 | 12/2008 | Sabata et al. | N/A | N/A |
| 11306648 | 12/2021 | Zhang | N/A | F02B 19/10 |
| 11359537 | 12/2021 | Gastaldi | N/A | F02F 1/40 |
| 11378042 | 12/2021 | Johnson | N/A | F02P 3/02 |
| 2001/0003281 | 12/2000 | Mori | 123/305 | F02F 1/242 |
| 2006/0137642 | 12/2005 | Artmann et al. | N/A | N/A |
| 2010/0183993 | 12/2009 | McAlister | 123/41.31 | F02D 41/401 |
| 2011/0048374 | 12/2010 | McAlister | 123/436 | F02M 57/06 |
| 2016/0356211 | 12/2015 | Wicks | N/A | F02D 41/0007 |
| 2017/0101923 | 12/2016 | Loetz | N/A | F02B 19/12 |
| 2018/0283235 | 12/2017 | Dalmas, II | N/A | F01M 11/0408 |
| 2018/0283264 | 12/2017 | Dalmas, II | N/A | F02F 3/28 |
| 2018/0283265 | 12/2017 | Dalmas, II | N/A | F01L 1/185 |
| 2018/0283311 | 12/2017 | Dalmas, II | N/A | F02B 33/30 |
| 2018/0283312 | 12/2017 | Dalmas, II | N/A | F02B 33/04 |
| 2020/0018254 | 12/2019 | Wicks | N/A | F02F 1/40 |
| 2020/0056535 | 12/2019 | Bouvy | N/A | F02B 47/02 |
| 2020/0173392 | 12/2019 | Brack | N/A | F02B 23/0642 |
| 2023/0100541 | 12/2022 | Browne | 123/1A | F02M 61/14 |

### FOREIGN PATENT DOCUMENTS

| Patent No. | Application Date | Country | CPC |
|---|---|---|---|
| 1207285 | 12/2001 | EP | N/A |
| 2017072031 | 12/2016 | JP | N/A |
| 9625592 | 12/1995 | WO | N/A |
| 1998053190 | 12/1997 | WO | N/A |
| 2000017505 | 12/1999 | WO | N/A |
| 2008041241 | 12/2007 | WO | N/A |
| 2022199841 | 12/2021 | WO | N/A |

### OTHER PUBLICATIONS

International Search Report issued in corresponding International Application No. PCT/US2024/026295; mailed Sep. 11, 2024 (6 pages). cited by applicant

Written Opinion of the International Searching Authority issued in corresponding International Application No. PCT/US2024/026295; dated Sep. 11, 2024 (8 pages). cited by applicant

---

*Primary Examiner:* Huynh; Hai H

*Assistant Examiner:* Hoang; Johnny H

*Attorney, Agent or Firm:* Osha Bergman Watanabe & Burton LLP

## Background/Summary

BACKGROUND

(1) Spark ignition engines generally use a pent roof combustion chamber, allowing larger valves than those typically used with a flat cylinder head. Diesel engines, in comparison, typically use a much stiffer flat cylinder head due to the higher peak pressures within the cylinder. Both geometries are able to be used for gas and hydrogen engines. In particular, a flat cylinder head offers the potential for higher combustion peak pressures when high performances are targeted. In such engines, it can be difficult to appropriately locate the fuel injector and the spark plug, which are preferably positioned in the center of the cylinder, without reducing the diameter of the intake and exhaust valves.

SUMMARY

(2) This summary is provided to introduce a selection of concepts that are further described below in the detailed description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

(3) In one aspect, embodiments disclosed herein relate to a hydrogen direct injection engine. The hydrogen direct injection engine may include a cylinder and a combustion chamber positioned at a center of the cylinder, where an axis of the combustion chamber is approximately identical to an axis of the cylinder. The hydrogen direct injection engine may also include one or more intake valves positioned around the cylinder, one or more exhaust valves positioned around the cylinder, a cylinder head mounted across a top surface of the cylinder, and a fuel injector positioned vertically at the center of the cylinder. The hydrogen direct injection engine may further include a piston disposed within the cylinder, and a spark plug positioned between the one or more intake valves.

(4) In another aspect, embodiments disclosed herein relate to a method. The method may include providing a hydrogen direct injection engine, where the hydrogen direct injection engine comprises a cylinder, a flat cylinder head sealing the cylinder, a combustion chamber positioned at a center of the cylinder between one or more intake valves and one or more exhaust valves, a fuel injector positioned at the center of the combustion chamber, and a spark plug flush mounted in the flat cylinder head. The method may also include injecting, during a compression stroke of the hydrogen direct injection engine, a volume of hydrogen into the combustion chamber using the fuel injector after a closure of an intake valve. The method may further include mixing a mixture of the volume of hydrogen and air within the combustion chamber and igniting the mixture using the spark plug.

(5) Other aspects and advantages of the claimed subject matter will be apparent from the following description and the appended claims.

## Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Specific embodiments of the disclosed technology will be described in detail with reference to the accompanying figures. Like elements in the various figures are denoted by like reference numerals for consistency. The size and relative positions of elements in the drawings are not necessarily drawn to scale. For example, the shapes of various elements and angles are not necessarily drawn to scale, and some of these elements may be arbitrarily enlarged and positioned to improve drawing legibility. Further, the particular shapes of the elements as drawn are not

necessarily intended to convey any information regarding the actual shape of the particular elements and have been solely selected for ease of recognition in the drawing.
(2) FIG. **1** shows a hydrogen direct injection engine in accordance with one or more embodiments.
(3) FIG. **2** shows a cylinder head in accordance with one or more embodiments.
(4) FIG. **3** shows a hydrogen direct injection engine in accordance with one or more embodiments.
(5) FIG. **4** shows a recessed piston head disposed in a combustion chamber in accordance with one or more embodiments.
(6) FIG. **5** shows a recessed piston head in accordance with one or more embodiments.
(7) FIG. **6** shows a computer system in accordance with one or more embodiments.
(8) FIG. **7** shows a flowchart of a method in accordance with one or more embodiments.
(9) FIG. **8** shows examples of combustion chamber profiles with different fuel injector recession depths in an engine according to embodiments of the present disclosure.

DETAILED DESCRIPTION

(10) In the following detailed description of embodiments of the disclosure, numerous specific details are set forth in order to provide a more thorough understanding of the disclosure. However, it will be apparent to one of ordinary skill in the art that the disclosure may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description.
(11) Throughout the application, ordinal numbers (e.g., first, second, third, etc.) may be used as an adjective for an element (i.e., any noun in the application). The use of ordinal numbers is not to imply or create any particular ordering of the elements nor to limit any element to being only a single element unless expressly disclosed, such as using the terms "before", "after", "single", and other such terminology. Rather, the use of ordinal numbers is to distinguish between the elements. By way of an example, a first element is distinct from a second element, and the first element may encompass more than one element and succeed (or precede) the second element in an ordering of elements.
(12) It is to be understood that the singular forms "a," "an," and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "a fracture" includes reference to one or more of such fractures.
(13) Terms such as "approximately," "substantially," etc., mean that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to those of skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.
(14) It is to be understood that one or more of the steps shown in the flowchart may be omitted, repeated, and/or performed in a different order than the order shown. Accordingly, the scope disclosed herein should not be considered limited to the specific arrangement of steps shown in the flowchart.
(15) Although multiple dependent claims are not introduced, it would be apparent to one of ordinary skill that the subject matter of the dependent claims of one or more embodiments may be combined with other dependent claims. Similarly, various features that are described in the context of a single embodiment (i.e., in combination) can be excised from the combination and may be directed to a sub-combination or variation of sub-combination.
(16) In the following description of FIGS. **1-6**, any component described with regard to a figure, in various embodiments disclosed herein, may be equivalent to one or more like-named components described with regard to any other figure. For brevity, descriptions of these components will not be repeated with regard to each figure. Thus, each and every embodiment of the components of each figure is incorporated by reference and assumed to be optionally present within every other figure having one or more like-named components. Additionally, in accordance with various embodiments disclosed herein, any description of the components of a figure is to be interpreted as

an optional embodiment which may be implemented in addition to, in conjunction with, or in place of the embodiments described with regard to a corresponding like-named component in any other figure.

(17) In one aspect, embodiments disclosed herein relate to a hydrogen direct injection engine configured to use hydrogen as a fuel. In particular, embodiments disclosed herein relate to a hydrogen direct injection engine including a flat cylinder head and a domed combustion chamber configured to allow appropriate placement of a fuel injector and a spark plug in a center region about a central axis of the combustion cylinder without requiring smaller intake and exhaust valves.

(18) FIG. **1** shows a hydrogen direct injection engine **100**, referred to as simply an "engine" throughout this application, in accordance with one or more embodiments. The engine **100** may include a cylinder **101** within an engine block **102**. In FIG. **1**, only a portion of the engine block **102** is shown, and only one cylinder **101** in the engine block **102** is shown, although an engine block **102** may have several cylinders. The cylinder **101** may include a combustion chamber **103** of the engine **100**. In accordance with one or more embodiments, the combustion chamber **103** may be dome-shaped, where the combustion chamber **103** cross-section is approximately equal or slightly less than the cross-section of the cylinder **101**. Additionally, a cylinder head **104** may be mounted at a top of the cylinder **101** and forms an upper end of the combustion chamber **103**. In one or more embodiments, the cylinder head **104** may be flat. For example, in the embodiment shown, the cylinder head **104** has a flat surface interfacing the combustion chamber **103**, where the flat surface may extend across the entire diameter of the combustion chamber. In one or more embodiments, as described in more detail below, the cylinder head may have a domed portion of the combustion chamber formed therein, extending inward from the flat surface, where the domed portion of the combustion chamber may be positioned along a central axis of the cylinder. In one or more embodiments, the volume of the combustion chamber **103** may be greater than 50% of the dead volume of the cylinder head **104**.

(19) A piston **105** may be arranged inside the cylinder **101** and forms a lower end of the combustion chamber **103**. In one or more embodiments, the piston **105** may have a recessed head. The piston **105** moves up and down inside the cylinder **101** during an engine cycle, and the volume of the combustion chamber **103** changes with the position of the piston **105**. Further, the piston **105** may be connected to a crankshaft (not shown) by a connecting rod. The crankshaft may convert the reciprocating motion of the piston **105** into rotary motion, as is well known in the art.

(20) A fuel injector **107** according to embodiments of the present disclosure may be mounted in the cylinder head **104**. A clamp (not pictured) may removably fix the fuel injector **107** to the cylinder head **104**. The clamp may be disposed on a top of the fuel injector **107** and be attached to the cylinder head **104** to maintain a position of the fuel injector **107**. The fuel injector **107** may be in fluid communication with the combustion chamber **103**. A spark plug **118** may be connected to and configured to interface with the combustion chamber **103**. The spark plug **118** may be used to ignite fuel within the combustion chamber **103**. In one or more embodiments, the spark plug **118** may be flush mounted in the cylinder head **104**, such that one or more electrodes (not pictured) of the spark plug **118** do not protrude into the combustion chamber **103**.

(21) As shown, the cylinder head **104** may include at least one intake passage **119**, which may fluidly communicate with the combustion chamber **103** via an intake valve **113**. Further, the cylinder head **104** may include at least one exhaust passage **111** fluidly connected to the combustion chamber **103** via an exhaust valve **114**. Although not shown, the combustion chamber **103** and the intake passage **119** may be connected to a source of air in a conventional manner. The air in the combustion chamber **103** and the intake passage **119** may be ambient air or a mixture of ambient air and recirculated exhaust gases. When the exhaust valve **114** is open, exhaust gases can be pushed out of the combustion chamber **103** into the exhaust passage **111**. An intake passage **119**, an exhaust passage **111** and associated components (e.g., valves **113**, **114** and fuel injector **107**) may be provided in the cylinder head **104** for each cylinder in the engine **100**, such as in the

arrangement shown in FIG. **1** for the cylinder **101**.
(22) In one or more embodiments, the fuel injector **107** may be used to directly inject fuel into the combustion chamber **103**. The fuel injector **107** may be fluidly connected to a fuel line **115**, which is in communication with a fuel supply **116**.
(23) In one or more embodiments, a computer **120** may include a control system, such as an engine control unit, which may control an opening and closing of the fuel injector **107** to deliver the fuel into the combustion chamber **103** at desired times during an engine cycle. The control system may also control opening and closing of the intake and exhaust valves **113**, **114**. The control system may further control the ignition timing of the engine (e.g., ignition timing of the spark plug **118** ignition based on the position of the piston, crankshaft, and/or intake/exhaust valves). In one or more embodiments, the computer **120** may include a processor and a user interface panel at which a user may provide an input, such as a command, to the computer **120**.
(24) In some embodiments, a cable (not shown), such as an electrical cable, may be coupled to the fuel injector **107**. The cable may provide power to the fuel injector **107** from a power source (not shown). Additionally, the cable may be connected to the computer **120** to control the fuel injector **107**. In one or more embodiments, the spark plug **118** may also be connected to the power source and computer to power and control timing of spark generation. The computer **120** may include instructions or commands to operate the fuel injector **107** and/or spark plug **118** automatically or a user may manually control the computer **120** at a user interface panel (not shown). It is further envisioned that the computer **120** may be connected to an office via a satellite such that a user may remotely monitor conditions and send commands to the fuel injector **107**. If leaks and performance issues are found, an alert may be sent to the control system to adjust or turn off the fuel injector **107** manually or automatically.
(25) The computer **120** may be, for example, a computer system **602**. FIG. **6** depicts a block diagram of a computer system **602** used to provide computational functionalities associated with described algorithms, methods, functions, processes, flows, and procedures as described in this disclosure, according to one or more embodiments. The illustrated computer **602** is intended to encompass any computing device such as a server, desktop computer, laptop/notebook computer, wireless data port, smart phone, personal data assistant (PDA), tablet computing device, one or more processors within these devices, or any other suitable processing device, including both physical or virtual instances (or both) of the computing device. Additionally, the computer **602** may include a computer that includes an input device, such as a keypad, keyboard, touch screen, or other device that can accept user information, and an output device that conveys information associated with the operation of the computer **602**, including digital data, visual, or audio information (or a combination of information), or a GUI.
(26) The computer **602** can serve in a role as a client, network component, a server, a database or other persistency, or any other component (or a combination of roles) of a computer system for performing the subject matter described in the instant disclosure. The illustrated computer **602** is communicably coupled with a network **630**. In some implementations, one or more components of the computer **602** may be configured to operate within environments, including cloud-computing-based, local, global, or other environment (or a combination of environments).
(27) At a high level, the computer **602** is an electronic computing device operable to receive, transmit, process, store, or manage data and information associated with the described subject matter. According to some implementations, the computer **602** may also include or be communicably coupled with an application server, e-mail server, web server, caching server, streaming data server, business intelligence (BI) server, or other server (or a combination of servers).
(28) The computer **602** can receive requests over network **630** from a client application (for example, executing on another computer **602**) and responding to the received requests by processing the said requests in an appropriate software application. In addition, requests may also

be sent to the computer **602** from internal users (for example, from a command console or by other appropriate access method), external or third-parties, other automated applications, as well as any other appropriate entities, individuals, systems, or computers.

(29) Each of the components of the computer **602** can communicate using a system bus **603**. In some implementations, any or all of the components of the computer **602**, both hardware or software (or a combination of hardware and software), may interface with each other or the interface **604** (or a combination of both) over the system bus **603** using an application programming interface (API) **612** or a service layer **613** (or a combination of the API **612** and service layer **613**. The API **612** may include specifications for routines, data structures, and object classes. The API **612** may be either computer-language independent or dependent and refer to a complete interface, a single function, or even a set of APIs. The service layer **613** provides software services to the computer **602** or other components (whether or not illustrated) that are communicably coupled to the computer **602**. The functionality of the computer **602** may be accessible for all service consumers using this service layer. Software services, such as those provided by the service layer **613**, provide reusable, defined business functionalities through a defined interface. For example, the interface may be software written in JAVA, C++, or other suitable language providing data in extensible markup language (XML) format or another suitable format. While illustrated as an integrated component of the computer **602**, alternative implementations may illustrate the API **612** or the service layer **613** as stand-alone components in relation to other components of the computer **602** or other components (whether or not illustrated) that are communicably coupled to the computer **602**. Moreover, any or all parts of the API **612** or the service layer **613** may be implemented as child or sub-modules of another software module, enterprise application, or hardware module without departing from the scope of this disclosure.

(30) The computer **602** includes an interface **604**. Although illustrated as a single interface **604** in FIG. **6**, two or more interfaces **604** may be used according to particular needs, desires, or particular implementations of the computer **602**. The interface **604** is used by the computer **602** for communicating with other systems in a distributed environment that are connected to the network **630**. Generally, the interface **604** includes logic encoded in software or hardware (or a combination of software and hardware) and operable to communicate with the network **630**. More specifically, the interface **604** may include software supporting one or more communication protocols associated with communications such that the network **630** or interface's hardware is operable to communicate physical signals within and outside of the illustrated computer **602**.

(31) The computer **602** includes at least one computer processor **605**. Although illustrated as a single computer processor **605** in FIG. **6**, two or more processors may be used according to particular needs, desires, or particular implementations of the computer **602**. Generally, the computer processor **605** executes instructions and manipulates data to perform the operations of the computer **602** and any machine learning networks, algorithms, methods, functions, processes, flows, and procedures as described in the instant disclosure.

(32) The computer **602** also includes a memory **606** that holds data for the computer **602** or other components (or a combination of both) that can be connected to the network **630**. For example, memory **606** can be a database storing data consistent with this disclosure. Although illustrated as a single memory **606** in FIG. **6**, two or more memories may be used according to particular needs, desires, or particular implementations of the computer **602** and the described functionality. While memory **606** is illustrated as an integral component of the computer **602**, in alternative implementations, memory **606** can be external to the computer **602**.

(33) The application **607** is an algorithmic software engine providing functionality according to particular needs, desires, or particular implementations of the computer **602**, particularly with respect to functionality described in this disclosure. For example, application **607** can serve as one or more components, modules, applications, etc. Further, although illustrated as a single application **607**, the application **607** may be implemented as multiple applications **607** on the computer **602**. In

addition, although illustrated as integral to the computer **602**, in alternative implementations, the application **607** can be external to the computer **602**.
(34) There may be any number of computers **602** associated with, or external to, a computer system containing a computer **602**, wherein each computer **602** communicates over network **630**. Further, the term "client," "user," and other appropriate terminology may be used interchangeably as appropriate without departing from the scope of this disclosure. Moreover, this disclosure contemplates that many users may use one computer **602**, or that one user may use multiple computers **602**.
(35) Turning now to FIG. **2**, FIG. **2** shows a cylinder head **104** in accordance with one or more embodiments. In one or more embodiments, the cylinder head **104** may be flat, with a domed portion of the combustion chamber **103** located centrally along a flat surface of the cylinder head **104**. Particularly, FIG. **2** shows an elevated view looking toward the flat surface of the cylinder head **104**. When the cylinder head **104** is mounted on a cylinder, the flat surface interfaces with an adjacent cylinder chamber. One or more intake valves **113** and one or more exhaust valves **114** may be arranged around the domed portion of the combustion chamber **103**, with the intake valves **113** and the exhaust valves **114** arranged parallel to one another about an axis **202**.
(36) FIG. **3** shows a cylinder head **104** of a hydrogen direct injection engine **100** in accordance with one or more embodiments. The domed portion of the combustion chamber **103** may be centrally located within the cylinder head **104**, with a central axis **312** of the combustion chamber **103** being approximately identical to the central axis of the cylinder **101**. The fuel injector **107** may be positioned vertically at or approximately along the central axis of the cylinder **101**. Specifically, the fuel injector **107** may be positioned in a recess in the cylinder head **104**, such that an injection tip of the fuel injector **107** interfaces with the domed portion of the combustion chamber **103**.
(37) In one or more embodiments, the tip of the fuel injector **107** may be recessed from an upper surface of the dome portion by a depth between 1 mm and 3 mm. For example, FIG. **8** shows examples of different combustion chamber profiles in an engine according to embodiments of the present disclosure. The combustion chamber profiles include a main combustion chamber **801** formed in the cylinder and a domed portion **802** of the combustion chamber formed in the cylinder head. An injection tip **803** of a fuel injector interfaces with the domed portion **802** along an upper surface **804** of the domed portion **802**. In a first configuration **810**, the injection tip **803** is positioned in-line, or flush, with the upper surface **804** of the domed portion **802** of the combustion chamber. In a second configuration **820**, the injection tip **803** is recessed a depth **805** of 2 mm from the upper surface **804** of the domed portion **802** of the combustion chamber. In a third configuration **830**, the injection tip **803** is recessed a depth **805** of 3 mm from the upper surface **804** of the domed portion **802** of the combustion chamber. As shown by the examples of the different recessed configurations, positioning the fuel injector in a recessed position from a domed portion of a combustion chamber may result in improved fuel jetting. When the tip of the fuel injector **107** is recessed, the jet flow may not be as attracted to the wall of the combustion chamber **103** (as according to the Coanda effect) and gas penetration is much improved. This may facilitate better mixing of the fuel with ambient air. In one or more embodiments, a fuel injector may be positioned axially within a fuel injector receptor at a selected axial position to position the injection tip of the fuel injector at a selected recess depth from the upper surface of the domed portion.
(38) The spark plug **118**, in accordance with one or more embodiments, may be positioned between the intake valves **113** to assist with cooling. The design of the spark plug **118** shown in FIG. **3** is not intended to be limiting. Other shapes or other configurations of electrode protrusions, such as surface or semi-surface electrodes, may be used without departing from the scope of this disclosure. The electrodes of the spark plug **118** may be located close to the wall of the domed portion of the combustion chamber **103** to promote cooling. In some embodiments, the ignition end of the spark plug may be flush with or recessed from the wall of the domed portion of the combustion chamber. The cylinder head **104** may be designed such that two or more water jackets

**302** are located proximate the spark plug **118** and in the vicinity of the tip of the fuel injector **107**.

(39) The domed portion of the combustion chamber **103** may be dome-shaped, with rounded upper corners. Each upper corner may have a radius of curvature **304**. In one or more embodiments, the radius of curvature **304** may be around 2 mm, which may allow for an acceptable thermomechanical stress and a close vicinity between the spark plug **118** and the fuel injector **107**. The domed portion of the combustion chamber **103** may have a height **306**, measured as an axial distance between the fire deck and the upper surface of the domed portion, and a cross-section **308**. In one or more embodiments, the fire deck may refer to the lower surface of the cylinder head **104**. As described herein, the fire deck, or lower surface, of the cylinder head **104** may be a flat surface. The cross-section **308** may be defined as the intersection of the combustion chamber **103** and the cylinder head **104** fire deck. The height **306** and the cross-section **308** may be selected in order to optimize the global compression ratio of the engine **100**, which may range between 9 and 13. The height **306** may be selected according to mechanical strength criteria and may be roughly independent of the engine bore size (the inner diameter of the cylinder). Further, the ratio between the cross-section **308** and the engine bore size may range between 3 and 4 (e.g., 3.5). For example, in some embodiments, the height **306** may be approximately 20 mm and the cross-section **308** may be approximately 25 mm.

(40) In one or more embodiments, the domed portion of the combustion chamber also has an upper cross-section measured between the side walls of the domed portion proximate the upper surface of the domed portion and parallel with the cross-section **308**. In one or more embodiments, the upper cross-section may be equal to or slightly less than the cross-section **308**, e.g., up to 10 percent less than the cross-section **308**.

(41) Distance **310** may refer to the smallest possible distance between the fire deck and the spark plug **118**. In one or more embodiments, distance **310** may be selected according to mechanical strength criteria. For example, in some embodiments, distance **310** may be approximately 6 mm.

(42) Each piston **105** may include a piston head, which may have a recessed area and may be composed of a material such as aluminum or steel. FIG. **4** shows a recessed piston head **400** disposed in a combustion chamber in accordance with one or more embodiment. In one or more embodiments, the piston head **400** may be recessed across the entire top surface, where the recess is uniform in shape. However, there are other embodiments, such as that shown in FIG. **5**, where the piston head **502** has a recess **504** which is non-uniform and non-symmetrical in shape. In the embodiment shown in FIG. **5**, the piston head **502** may be rotationally oriented in a cylinder such that the portion with a smaller volume recess area **506** aligns with an exhaust side of the piston head (e.g., closer to exhaust valves interfacing the cylinder), and the portion with a larger volume recess area **508** aligns with an intake side of the piston head (e.g., closer to intake valves interfacing the cylinder).

(43) Turning back to FIG. **4**, the recess of the piston may have a radius of curvature **405** and a recess depth **406**. In one or more embodiments, the radius of curvature **405** may be approximately 5 mm to avoid any hot spots on the surface of the piston head **400**. The recess depth **406** may be selected according to a chosen compression ratio. In one or more embodiments, the recess depth **406** may range between 5 mm and 10 mm. The shallow shape of the piston may allow for the flame within the cylinder **101** to freely propagate towards the wall of the cylinder **101**, while limiting wall heat losses. The squish height **402** may refer to the distance between the piston **105** and the cylinder head **104** fire deck when the piston **105** is at a top dead center (TDC) position. In one or more embodiments, the squish height **402** may be between 2 mm and 3 mm, which is significantly higher than the squish height of conventional SI engines. A larger squish height **402** may avoid pre-ignition or knock with hydrogen.

(44) The squish width **404** may refer to a portion of the piston head which is parallel to the flat surface of the cylinder head **104**. In one or more embodiments, the squish width **404** may range from 2 to 4 mm (e.g., approximately 3 mm) and may be selected as the minimum value which may

secure the desired thermomechanical stress.
(45) In one or more embodiments, cylinder heads **104** having a domed portion of a combustion chamber **103** formed therein, such as shown in FIG. **3**, may be used in combination with a cylinder **101** having a recessed piston head, such as shown in FIG. **4** or in FIG. **5**.
(46) FIG. **7** depicts a flowchart in accordance with one or more embodiments. More specifically, FIG. **7** depicts a flowchart **700** of a method of calculating an average total organic carbon value corresponding to the net source rock thickness. Further, one or more blocks in FIG. **7** may be performed by one or more components as described in FIGS. **1-6**. While the various blocks in FIG. **7** are presented and described sequentially, one of ordinary skill in the art will appreciate that some or all of the blocks may be executed in different orders, may be combined, may be omitted, and some or all of the blocks may be executed in parallel. Furthermore, the blocks may be performed actively or passively.
(47) Initially, a hydrogen direct injection engine **100** may be provided, S**702**. The hydrogen direct injection engine **100** may be operated in a similar manner to a conventional single injection engine in terms of operating point (torque and speed) and tuning, with an optimization of the injection and ignition timing. The optimization procedure may differ for hydrogen direct injection engines **100** due to the properties of the hydrogen-air mixture. For example, the ignition may occur later in the cycle, and combustion may be more sensitive to pre-ignition necessitating a good mixture homogeneity and a good cooling of the combustion chamber **103**.
(48) The engine **100** may include a cylinder **101**, a flat cylinder head **104** sealing the cylinder **101**, and a combustion chamber **103** positioned centrally in the cylinder **101**. A domed portion of the combustion chamber **103** formed in the cylinder head **104**, in accordance with one or more embodiments, may be positioned between one or more intake valves **113** and one or more exhaust valves **114**. The engine **100** may also include a fuel injector **107** positioned along a central axis of the combustion chamber **103** and a spark plug **118** flush mounted in the flat cylinder head **104**.
(49) During a compression stroke of the engine **100**, a volume of hydrogen may be injected into the combustion chamber **103** using the fuel injector **107**, S**704**. In one or more embodiments, injection of the volume of hydrogen may occur after the intake valve **113** has closed. After injection, the volume of hydrogen may be mixed with air within the combustion chamber **103** to form a mixture, S**706**. The mixture may then be ignited using the spark plug **118**, S**708**.
(50) In one or more embodiments, movement of the recessed piston head **400** through the combustion chamber **103** may induce free propagation of the flame produced by the spark plug **118** towards the wall of the cylinder **101**. This may limit heat losses from the wall of the cylinder **101**.
(51) In one or more embodiments, a second volume of hydrogen may be injected into the combustion chamber **103** during an intake stroke of the engine **100**. Specifically, the second injection may occur when the intake valve **113** lift is at a maximum. This may enhance mixing between the hydrogen and the ambient air in the combustion chamber **103** by taking advantage of the high air velocities which are inherent at maximum intake valve **113** lift.
(52) Embodiments of the present disclosure may provide at least one of the following advantages. While conventional single injection combustion systems are generally used for hydrogen, these conventional systems take poor advantage of the very high flame velocity and the wide range of ignitability in terms of air/fuel ratio, for both lean and rich mixtures. Due to geometric constraints, the spark plug is typically located far from the tip of the injector, leading to a high air/fuel dispersion at the ignition timing in the vicinity of the spark plug. Further, conventional systems rely on high air motion generated by the intake passages, with high wall losses and a sensitivity to industrial dispersion.
(53) In contrast, the present invention does not rely on any bulk air motion, instead relying on injection strategy to sufficiently mix the air fuel mixture within the combustion chamber. The intake passages and valves may therefore be selected to maximize air mass flow into the combustion chamber, which may maximize engine performance. The present invention also allows

for both the fuel injector and the spark plug to be located in the center of the cylinder, reducing the time for the flame front to reach the cylinder wall. Further, the positioning of the fuel injector and the spark plug in the center of the cylinder does not require reducing the diameters of either the intake valves or the exhaust valves. Given the close vicinity of the injector tip and the spark plug, the risk of misfire is greatly reduced. Since the fuel injector is positioned vertically in the center of the combustion chamber, the fuel injector generates a fuel jet which can easily be mixed with air flow during the intake stroke. The present invention also allows for the application of a variety of injection strategies (e.g., fully premixed, partially premixed, stratification) according to engine load. In general, the present invention allows for both a high power density and a stable, efficient combustion at low loads.

(54) Although only a few example embodiments have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the example embodiments without materially departing from this invention. Accordingly, all such modifications are intended to be included within the scope of this disclosure as defined in the following claims.

## Claims

1. A hydrogen direct injection engine, comprising: a cylinder; a combustion chamber positioned at a center of the cylinder, the combustion chamber having a domed portion; one or more intake valves in communication with the combustion chamber; one or more exhaust valves in communication with the combustion chamber; a cylinder head mounted across a top surface of the cylinder; a fuel injector positioned vertically along a central axis of the cylinder; a piston disposed within the cylinder; and a spark plug connected to and interfacing with the combustion chamber, wherein the domed portion of the combustion chamber is recessed within the cylinder head and positioned between the one or more intake valves and the one or more exhaust valves, wherein the fuel injector comprises an injection tip interfacing the domed portion and positioned at a recessed depth from an upper surface of the domed portion of the combustion chamber.
2. The hydrogen direct injection engine of claim 1, wherein an axis of the combustion chamber is coaxial with the central axis of the cylinder.
3. The hydrogen direct injection engine of claim 1, further comprising two water jackets positioned proximate the spark plug and the injection tip of the fuel injector.
4. The hydrogen direct injection engine of claim 1, wherein the spark plug is flush mounted in the cylinder head such that one or more electrodes included in the spark plug do not protrude into the combustion chamber.
5. The hydrogen direct injection engine of claim 1, wherein the cylinder head has a flat fire deck surrounding the domed portion, and wherein the one or more intake valves and the one or more exhaust valves are arranged within the flat fire deck.
6. The hydrogen direct injection engine of claim 1, wherein the piston has a radius of curvature equal to 5 mm.
7. The hydrogen direct injection engine of claim 1, wherein a squish height of the cylinder may be in a range of 2 mm to 3 mm.
8. The hydrogen direct injection engine of claim 1, wherein the domed portion has a radius of curvature equal to 2 mm.
9. The hydrogen direct injection engine of claim 1, wherein the domed portion comprises: a height measured as an axial distance between the upper surface of the domed portion and a fire deck portion of the cylinder head surrounding the domed portion; and a cross-section measured across a base of the domed portion between the fire deck portion.
10. The hydrogen direct injection engine of claim 1, wherein the piston has a recessed area with a depth selected according to a chosen compression ratio.
11. The hydrogen direct injection engine of claim 10, wherein the depth is in a range of 5 mm to 10

mm.

12. The hydrogen direct injection engine of claim 10, wherein the recessed area comprises a first volume recess area and a second volume recess area, wherein the first volume recess area is smaller than the second volume recess area, and wherein the piston is rotationally oriented in the cylinder such that first volume recess area aligns with an exhaust side of the cylinder, and the second volume recess area aligns with an intake side of the cylinder.

13. The hydrogen direct injection engine of claim 1, wherein a volume of the combustion chamber is greater than 50% of a dead volume of the cylinder head.

14. A hydrogen direct injection engine, comprising: a cylinder; a cylinder head mounted across a top surface of the cylinder; a combustion chamber provided in the cylinder, an intake valve positioned around the cylinder head and interfacing the combustion chamber; an exhaust valve positioned around the cylinder head and interfacing the combustion chamber; a domed portion of the combustion chamber recessed within the cylinder head and positioned between the intake valve and the exhaust valve; a fuel injector positioned vertically along a central axis of the cylinder; a piston disposed within the cylinder; and a spark plug interfacing the domed portion of the combustion chamber, wherein the fuel injector comprises an injection tip interfacing the domed portion and positioned at a recessed depth from an upper surface of the domed portion of the combustion chamber.

15. A method, comprising: providing a hydrogen direct injection engine, wherein the hydrogen direct injection engine comprises: a cylinder, a cylinder head with a flat fire deck sealing the cylinder, a domed combustion chamber recessed within a center of the cylinder head and positioned between one or more intake valves and one or more exhaust valves, a fuel injector positioned at the center of the domed combustion chamber, and a spark plug flush mounted in the cylinder head, wherein the fuel injector comprises an injection tip interfacing the domed portion and positioned at a recessed depth from an upper surface of the domed combustion chamber; injecting, during a compression stroke of the hydrogen direct injection engine, a volume of hydrogen into the domed combustion chamber using the fuel injector after a closure of an intake valve; mixing a mixture of the volume of hydrogen and air within the domed combustion chamber; and igniting the mixture using the spark plug.

16. The method of claim 15, further comprising moving a recessed piston head through the domed combustion chamber and inducing free propagation of a flame produced by the spark plug towards a wall of the cylinder.

17. The method of claim 16, further comprising limiting heat losses from the wall of the cylinder.

18. The method of claim 15, further comprising injecting, during an intake stroke of the hydrogen direct injection engine, a second volume of hydrogen into the domed combustion chamber.